FUNCTIONAL SPECIFICATION

Vendor Master Interface

Legacy Procurement Hub to SAP S/4HANA

RICEFW object	Interface (I)
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Prepared for	SAP S/4HANA Implementation Program

Purpose

This sample specification defines the inbound interface that creates and changes supplier business partners in SAP S/4HANA from the legacy Procurement Hub. It is intended as a reusable project template; organization codes, endpoints, and field rules must be confirmed during design.

Confidential - Sample implementation artifact

1. Business Overview

The Procurement Hub remains the system of record for vendor onboarding and maintenance during the transition period. Approved vendor records are sent to SAP S/4HANA so that purchase orders, invoices, and payments can be processed against a consistent supplier master. In S/4HANA, suppliers are maintained as Business Partners with supplier roles.

1.1 Scope

In scope	Out of scope
Create and change supplier general, company code, purchasing, address, bank, tax, and contact data.	Vendor deactivation, duplicate cleansing, workflow approval, and payment execution.
Synchronous validation response and nightly reconciliation report.	Historical vendor migration and non-supplier business partner categories.

1.2 Assumptions and Dependencies

ID	Assumption / dependency	Owner
A-01	Legacy vendor ID is stable and is stored in SAP as an external BP identification.	Data Governance
A-02	Company code, purchasing organization, payment terms, reconciliation account, and tax code mapping are approved before SIT.	Finance / Procurement
A-03	SAP Integration Suite routes messages and applies mutual TLS authentication.	Integration
A-04	SAP MDG is not in scope for this initial release; changes originate in the legacy system.	Program
A-05	Bank details are sent only for vendors authorized by the legacy approval workflow.	Treasury

1.3 Success Criteria

A valid vendor is created or updated within 15 minutes of approval; malformed messages are rejected without partial master-data updates; and daily source-to-target counts reconcile to 100 percent after approved reprocessing.

2. Interface Design

Attribute	Design
Interface name	LEGACY_VENDOR_TO_S4_BP
Direction	Inbound to SAP S/4HANA
Trigger / frequency	Near real-time event after vendor approval or change; scheduled reconciliation at 02:00 local time.
Transport	HTTPS REST through SAP Integration Suite; JSON UTF-8 payload; mutual TLS and OAuth 2.0 client credentials.
SAP processing	Integration flow validates/enriches, then invokes the SAP Business Partner supplier API / inbound service.
Idempotency	SourceVendorId + SourceChangeTimestamp. Replayed messages with an unchanged key are acknowledged without duplicate creation.
Volume	Estimated 300 creates/changes per business day; peak 50 messages per 15 minutes.
Retention	Payload and application logs retained 90 days; bank values masked in operational logs.

2.1 Processing Flow

1. Legacy approval publishes VendorUpsert. 2. Integration Suite authenticates, validates schema, and translates codes. 3. SAP determines the existing Business Partner by external legacy ID. 4. SAP creates or updates BP general data, supplier company code, and purchasing views in one logical unit. 5. A response returns the BP number, processing status, and any business errors. 6. Failed records enter the support queue for correction and replay.

2.2 Key Business Rules

Rule	Description
BR-01	Legal name, country, language, and supplier account group/category are mandatory for create.
BR-02	If SourceVendorId already exists, update the related Business Partner; do not create a new BP.
BR-03	Company code data is required when payment terms, reconciliation account, or withholding-tax data is supplied.
BR-04	Bank account changes require a source approval flag = Y; otherwise reject with error VM-120.
BR-05	ISO country and currency codes must be valid; unmapped legacy codes are rejected.
BR-06	Blank source values do not clear SAP values unless ClearIndicator = Y for the specific field group.

3. Field Mapping

The mapping below represents the minimum viable supplier payload. Final SAP API property names and controlled values are to be confirmed in the interface technical design.

Legacy JSON field	SAP S/4HANA target	Format	Req.	Transformation / validation
Header.sourceVendorId	BP Identification (LEGACY_VENDOR)	Char 20	Y	External unique key; immutable.
Header.changeTimestamp	Extension: Source Change Timestamp	ISO 8601	Y	Used for idempotency and audit.
General.legalName	BP Organization Name 1	Char 80	Y	Trim and collapse multiple spaces.
General.searchTerm	BP Search Term 1	Char 20	N	Default to first 20 chars of legal name.
General.language	BP Correspondence Language	ISO 639-1	Y	Map legacy language code.
Address.country	BP Address Country	ISO 3166-1	Y	Validate against SAP country table.
Address.city / postalCode	BP Address City / Postal Code	Char 40 / 10	Y	Postal code format by country.
Tax.taxNumber	BP Tax Number 1	Char 30	Conditional	Mandatory where country tax registration requires it.
Company.companyCode	Supplier Company Code	Char 4	Y	Must be active SAP company code.
Company.reconAccount	Supplier Reconciliation Account	Char 10	Y	Map legacy liability class.
Company.paymentTerms	Supplier Payment Terms	Char 4	Y	Map approved payment term.
Purchasing.purchOrg	Supplier Purchasing Organization	Char 4	Y	At least one purchasing organization.
Purchasing.orderCurrency	Supplier Order Currency	ISO 4217	N	Default from purchasing organization.
Bank.iban / accountNo	BP Bank Detail	Char 34 / 18	Conditional	One is required when bank data is sent.

Sample Payload


```
{"sourceVendorId":"L-104582","changeTimestamp":"2026-10-18T14:30:00Z","legalName":"Acme Industrial Supply LLC","companyCode":"1000","purchOrg":"1000","paymentTerms":"0001"}
```

4. Error Handling and Operations

Category	Handling	Response / ownership
Schema or security failure	Reject before SAP invocation; log correlation ID and error code.	HTTP 400/401/403; legacy integration support.
Business validation failure	Do not update SAP. Place message in exception queue with field-level detail.	HTTP 422; master-data analyst corrects source or mapping.
SAP technical failure	Retry automatically up to 3 times with exponential backoff. Queue after final failure.	HTTP 500/503; SAP basis / integration support.
Partial processing	Not permitted. General, company code, purchasing, and bank updates execute atomically where supported.	Return VM-500; manual review.
Duplicate / replay	Compare idempotency key and last applied timestamp.	HTTP 200 with status DUPLICATE_IGNORED.

4.1 Error Codes

Code	Meaning	Corrective action
VM-101	Required legal name is missing.	Populate General.legalName and replay.
VM-115	Company code is not mapped or inactive.	Correct mapping or activate valid company code.
VM-120	Bank update lacks approval flag.	Obtain approval in source and resend.
VM-130	Tax number format is invalid for country.	Correct tax registration data.
VM-500	Unexpected SAP processing error.	Review correlation ID and SAP application log.

4.2 Monitoring and Reconciliation

Integration monitoring alerts the support team for messages in error longer than 30 minutes. A daily reconciliation compares approved source changes, successful SAP responses, failed messages, and queued retries by date and source vendor ID. Any unmatched record is investigated within one business day.

4.3 Security and Controls

Transport uses TLS 1.2 or higher. Service credentials are held in the secure credential store. Access to payload monitoring is limited to integration support and authorized master-data users. Operational logs mask bank account and tax-number values. All create/change events retain the source system ID, correlation ID, timestamp, and SAP user/service ID.

5. Test Scenarios and Acceptance

ID	Scenario	Expected result
T-01	Create a domestic supplier with one company code and purchasing organization.	BP and supplier roles created; response contains SAP BP number.
T-02	Update an existing supplier address and payment terms.	Existing BP updated; no duplicate BP created.
T-03	Send a missing legal name.	Message rejected with VM-101; SAP unchanged.
T-04	Send bank data without source approval.	Message rejected with VM-120; bank details unchanged.
T-05	Replay a successfully processed message.	Acknowledged as DUPLICATE_IGNORED; no duplicate update.
T-06	Simulate SAP unavailable.	Automatic retry occurs; message queues after third failure.
T-07	Run daily reconciliation with one deliberately failed record.	Report identifies record and its correlation ID.
T-08	Validate role-based log access.	Unauthorized user cannot view unmasked payload data.

5.1 Acceptance Criteria

Business and IT approve the interface when all critical scenarios pass, mappings are signed off, no high-severity security findings remain, and reconciliation demonstrates complete traceability from source approval through SAP processing.

6. Approvals

Role	Name	Approval	Date
Business Process Owner	_____	_____	_____
Finance Data Owner	_____	_____	_____
SAP Functional Lead	_____	_____	_____
Integration Lead	_____	_____	_____

Document Change Log

Version	Date	Author	Description
1.0	2026-10-18	Implementation Team	Initial sample functional specification.